

EquationWright

v7.3 Complete User Manual

Ultimate Math Worksheet Builder

What this manual is for

A plain-language guide to creating worksheets and quizzes, using every major student and teacher feature, saving and sharing work, reviewing results, using Smart Review and AI tools, working offline, using advanced replay tools, rebuilding the application from source, and extending the modular source safely.

Final approved v7.3 release

appVersion: v7.3_full_catalog_protocols

81 units • 424 generator families • 317 unit-scoped topics

Online and offline editions share the same assessment engine.

Prepared from the final approved v7.3 application, modular source, curriculum profile, release notes, and public protocol schemas.

Contents

Section	What you will learn
1	Start here: the 5-minute quick start
2	What EquationWright does
3	Online vs. offline editions
4	Main screen: what every control means
5	Creating a worksheet
6	Creating and taking a quiz
7	Seeds, repeatability, variety, and topics
8	Teacher mode and Student mode
9	Presets and common classroom workflows
10	Printing, PDF, LaTeX, and scratch work
11	Blueprints, saved blueprints, and share links
12	Smart Review and personal practice history
13	Class tools and result collection
14	AI tutor and local AI
15	Advanced protocol, replay, diagnostics, and profiles
16	Self-test and troubleshooting
17	Ways to use EquationWright
18	Rebuilding from source
Appendix A	Course and unit catalog
Appendix B	Quick reference
Appendix C	Developer notes: build, register, and add new content

1. Start here: the 5-minute quick start

EquationWright is a single-file math practice system. You open the HTML file in a browser, choose what you want to practice, and generate a reproducible worksheet or interactive quiz.

Quick start for a teacher

1. **Open the app.** Use the online HTML for normal connected use or the offline HTML when you want the math renderer bundled into the file.
2. **Choose a course.** Pick a Curriculum profile, then choose a Mode such as Calculus I, Statistics I, Differential Equations, or Mixed.
3. **Set the seed.** Type a memorable name such as MTH263-Quiz-3. The same seed plus the same settings reproduces the same assessment.
4. **Choose question counts and topics.** Enter a number for each unit. Use Topics to include only the concepts you want.
5. **Choose worksheet or quiz behavior.** Turn Quiz Mode on for interactive multiple choice. Turn Answer Key and Worked Solutions on or off as needed.
6. **Generate.** Click Generate worksheet / Generate quiz.
7. **Deliver it.** Print or Save PDF, export a blueprint, save it locally, copy a share link, or create a locked Student link.

Quick start for a student

8. **Switch to Student mode.** Student mode forces interactive Quiz Mode and hides answer pages and teacher-only tools.
9. **Choose or open an assigned quiz.** A teacher can give you a share link, or you can use a preset such as Quick Quiz (5).
10. **Answer the questions.** Click an answer choice. The worked solution becomes available after you answer.
11. **Review.** Use Review missed questions, Retake this quiz, New quiz like this, or Start Smart Review (local).
12. **Keep or share results.** Export CSV/JSON, submit to a configured collector, or import result JSON files later into Personal practice history.

The fastest no-setup practice

One-click practice

Choose Student → Quick Quiz (5). EquationWright turns on Quiz Mode, turns off answer/solution pages, distributes five questions across the selected course, and generates the quiz immediately.

2. What EquationWright does

EquationWright combines a deterministic math generator, worksheet/quiz renderer, student practice tools, result analysis, and portable replay formats in one browser-based application.

Feature	What it means
Generate math practice	81 units and 424 published generator families across the full catalog.
Filter by course/unit/topic	Choose a course, set per-unit counts, and restrict individual units to selected topics.
Reproduce an assessment	Same seed + same authoritative settings recreates the same assessment within a compatible engine.
Worksheet mode	Creates printable questions with optional answer key and worked-solution pages.
Quiz mode	Interactive multiple-choice practice with intentional 2-, 3-, or 4-choice items as supported by answer capacity.
Smart Review	Builds a fresh local remediation/challenge quiz from the learner's actual results.
Scratch Pad	Floating multi-page digital scratch paper with pen, eraser, line, colors, paper styles, and PNG export.
Save/share	Saved blueprint library, portable blueprint JSON, and share links.
History and class analysis	Import result JSON files locally for learner trends or class summaries.
AI-assisted tutoring	Optional bring-your-own-provider analysis; the app still generates and grades the math.
Offline use	Offline edition embeds pinned MathJax 3.2.2 for formula rendering without a network.
Advanced replay	Archival blueprints, strict replay, verify-current checks, capabilities export, diagnostics, and curriculum profiles.

Important distinction

The AI is optional. EquationWright's math generation and grading are local/deterministic. AI tools read labeled result information and suggest a drill; the application creates and grades that drill.

3. Online vs. offline editions

Edition	Use it when...	Math rendering
EquationWright_v7_3_ONLINE.html	You normally have internet access and want the smaller file.	Loads MathJax from a CDN or a custom/local path.
EquationWright_v7_3_OFFLINE.html	You want the entire math renderer inside the app file.	Pinned MathJax 3.2.2 is embedded; no CDN is required for formulas.

The two editions are packaging variants of the same assessment engine. They share the same app version, assessment compatibility identity, catalog digest, and determinism protocol.

What “offline” does and does not mean

The offline edition removes the network dependency for math rendering. Features that intentionally contact an external service—such as a hosted AI provider or a remote results collector—still need access to that service. A local OpenAI-compatible AI endpoint can be used without a hosted provider.

4. Main screen: what every control means

Top settings

Control	What it does
Worksheet seed	The reproducibility key. Use a meaningful seed when you want to recreate the same assessment.
Curriculum profile	Controls course presentation, labels, ordering, defaults, and crosswalk information.
Mode	Selects the course or Mixed catalog view.
Quiz Mode	Changes the generated assessment to interactive multiple-choice practice.
Answer Key	Adds an answer-only page to worksheet output.
Worked Solutions	Adds solution-outline pages to worksheet output.
Variety	Low / Medium / High. Higher settings try more candidate problems per slot and choose more varied ones.
Coverage pills	Shows live coverage information for the selected mode—units, generators, and topics.

Control	What it does
Unit count	How many questions to create from that unit. Use 0 to omit the unit.
Topics	Restricts a unit to selected concepts.
Total Qs	Shows the current total question count and warnings.

Action buttons

Button/area	Use
Teacher / Student	Switches the user interface and safety defaults.
Presets	Quick Quiz (5), Homework Practice (15), Unit Review (20), Teacher Copy, Student Copy.
Generate	Re-renders the assessment from current settings.
Print / Save PDF	Opens the browser print workflow.
Export LaTeX	Downloads a LaTeX version of the current assessment.
Export / Import Blueprint	Portable settings-based assessment files.
Saved blueprints	Browser-local blueprint library.
Copy share link	Copies a URL containing a routine blueprint so another browser can recreate the assessment.
Randomize seed	Creates a new random-looking seed and regenerates.

5. Creating a worksheet

A reliable worksheet workflow

- Choose the Curriculum profile and Mode.** For the built-in VCCS/NVCC profile, the Mode list contains the supported mathematics courses plus Mixed.
- Enter a seed.** Use a name that tells you what it belongs to: for example MTH264-Week5-Practice.
- Set unit counts.** Each visible unit has a number field. Set unwanted units to 0.
- Optional: filter topics.** Open Topics under a unit, leave only the topics you want, then choose Done.
- Set Variety.** Medium is a useful default; High asks the selector to work harder to avoid near-duplicate structure.

18. **Leave Quiz Mode off.** This produces standard worksheet cards and work space.
19. **Choose Answer Key and Worked Solutions.** Turn them on for teacher copies; turn them off for a clean student handout.
20. **Click Generate worksheet.** The output appears below the controls.
21. **Print / Save PDF or Export LaTeX.** Use the format that fits your class workflow.

Topic filter behavior

If you clear every topic in a unit, the interface reports “None (will fall back to all).” That does not disable the unit. To remove a unit completely, set its question count to 0.

Teacher Copy and Student Copy presets

These two presets are convenience switches that keep your current counts:

- **Teacher Copy:** worksheet mode, answer key on, worked solutions on.
- **Student Copy:** worksheet mode, answer key off, worked solutions off.

Auto-saved working state

The current seed, mode, variety, page toggles, counts, and topic selections are saved in browser local storage when available. When you return in the same browser context, EquationWright can restore that working state.

6. Creating and taking a quiz

Create a quiz

22. **Turn on Quiz Mode,** or use Student → Quick Quiz (5).
23. **Turn off Answer Key and Worked Solutions for student delivery.** Student mode enforces these choices automatically.
24. **Set counts/topics and seed, then Generate quiz.** The questions become interactive multiple-choice items.
25. **Share or use the quiz locally.** A share link reproduces the same routine assessment settings.

Take a quiz

Each multiple-choice question shows answer buttons. When a learner selects an answer, EquationWright scores it and exposes the worked solution for that question.

- **Score bar:** shows the current score.
- **Answered count:** shows progress through the quiz.
- **Worked solution:** unlocks after the learner answers that question.
- **Choice count:** may be 2, 3, or 4 depending on what that generator can support reliably.

Quiz action bar

Action	What happens
Review missed questions	Hides correct items so the learner can focus on missed questions; toggles back to Show all questions.
Start Smart Review (local)	Builds a fresh, targeted local review quiz from the current results.
Retake this quiz	Regenerates with the same seed/settings, so the quiz is the same.
New quiz like this	Changes the seed and generates a fresh assessment with the same general settings.
Export results (CSV)	Creates a human-friendly spreadsheet-style result file.
Export results (JSON)	Creates the full EquationWright result file used by history/class tools.
Submit results	Posts a privacy-stripped result to a configured collector.
Ask AI tutor	Sends the appropriate result summary to the selected AI provider and can return a prescribed drill.

7. Seeds, repeatability, variety, and topics

Seeds: the core reproducibility idea

A seed is a text value used by the deterministic generator. The same normalized seed and the same authoritative assessment settings reproduce the same assessed content in a compatible EquationWright engine.

Remember

The seed is not the only setting. Mode/course, counts, topic restrictions, quiz behavior, variety, and compatibility identity matter too. For portable routine replay, use a blueprint or share link instead of writing down only the seed.

Randomize seed

Randomize seed replaces the seed with a new “Seed-...” value and regenerates. Use it when you want another version of the same general practice setup.

Variety

Setting	Selector behavior	Good use
Low	Tries fewer candidates for each slot.	Fast/basic generation.
Medium	Tries more candidates and balances variety.	Normal default.
High	Tries the most candidates and favors the most different result.	Longer review sets or when you want maximum structural variety.

Topic filters

26. **Click Topics under a unit.** A modal shows that unit's live topic list.
27. **Uncheck topics you do not want.** The summary pills update as you change the selection.
28. **Use Select all to remove restrictions.** An empty internal restriction means "all topics."
29. **Use Clear only when you want to reset the selection UI.** Because a zero-topic restriction falls back safely to all, use the unit count field to exclude a whole unit.
30. **Click Done.** Your selection is saved locally with the other worksheet settings.

8. Teacher mode and Student mode

Mode	What is available
Teacher	All worksheet/quiz controls, answer pages, worked solutions, blueprints, class tools, advanced protocol/replay tools, LaTeX, Self-test, and other authoring features.
Student	Quiz-focused interface. Quiz Mode is forced on; Answer Key and Worked Solutions are forced off. Teacher-only controls are hidden. Personal history, scratch pad, and student-facing review tools remain available.

Locked Student links

A teacher can enter a Student-lock PIN and choose Copy locked student link. The link opens in Student mode. Switching to Teacher mode requires the PIN for that session.

Not exam security

The built-in Student-lock PIN is explicitly a deterrent to casual peeking. It is not secure examination software and should not be treated as high-stakes access control.

9. Presets and common classroom workflows

Preset	What it sets
Quick Quiz (5)	Quiz on; answer/solution pages off; exactly 5 total questions distributed across the visible course.
Homework Practice (15)	Worksheet; answer key and worked solutions on; 15 total questions.
Unit Review (20)	Worksheet; answer key on; worked solutions off; 20 total questions.
Teacher Copy	Keeps current counts; worksheet; answer key + worked solutions on.
Student Copy	Keeps current counts; worksheet; answer key + worked solutions off.

How Mixed mode distributes presets

For Mixed mode, preset totals are distributed with a seeded course round-robin. A short Mixed quiz therefore samples multiple course families reproducibly rather than simply filling the first units in the catalog.

Example: one quiz, many students

31. **Create the quiz with a memorable seed.**
32. **Set Student mode or disable teacher pages.**
33. **Choose Copy share link or Copy locked student link.**
34. **Give every student the same link.** They receive the same assessment settings on their own device.
35. **Collect JSON results or use a teacher-owned collector.**

Example: parallel versions

36. **Create Version A with a meaningful seed.** Example: MTH263-Quiz4-A.
37. **Use New quiz like this or change the seed.** Keep the course/count/topic settings unchanged.
38. **Save both blueprints.** This gives you two portable, clearly named versions.

10. Printing, PDF, LaTeX, and scratch work

Print / Save PDF

Print / Save PDF uses the browser's print dialog. The print stylesheet removes the control panel and formats the worksheet pages for paper. Whether answer and solution pages are included is controlled by the Answer Key and Worked Solutions toggles.

LaTeX export

- **Keep exact fractions:** keeps exact rational forms instead of decimalizing values such as $\frac{1}{2}$ where the exporter supports it.
- **Diagrams — Remove:** omits diagrams from LaTeX output.
- **Diagrams — Replace with placeholder:** inserts a placeholder where a diagram would have appeared.

Floating Scratch Pad

The scratch pad is separate from grading. It is digital scrap paper for working problems, not a solver.

Control	Use
Open	Click  Scratch Pad.
Pen / Eraser / Line	Draw freehand, erase strokes, or draw a straight line.
Undo / Redo	Step backward or forward through scratch-pad changes.
Ink colors	Black, blue, red, and green.
Stroke	Adjust line thickness. Eraser uses a wider effective stroke.
Paper	Plain, Graph, Coordinate, or Dots.
Workspace size	Auto, Small, Medium, or Large.
Pages	Up to 5 scratch pages; use previous/next, + Page, and Delete page.
Clear	Clears the current page; the clear action can be undone.
Save PNG	Downloads the current scratch page as an image including the selected paper background.
Minimize / Close	Minimize keeps it compact; Close hides it for the session.
Pop out	Opens a simpler separate scratch-pad window on browsers that allow pop-ups.

Scratch-pad keyboard shortcuts

Shortcut	Action
P	Pen
L	Line

Shortcut	Action
E	Eraser
G	Cycle paper style
Esc	Minimize / restore
Ctrl/Cmd + Z	Undo
Ctrl/Cmd + Shift + Z	Redo

Scratch pad storage

Scratch work is not part of the quiz result, seed, or grading record. Save important scratch work as PNG before closing/reloading if you want to keep it.

11. Blueprints, saved blueprints, and share links

What is a blueprint?

A routine v7.3 blueprint is a portable JSON description of the authoritative assessment settings. It stores the seed, positive unit counts/order, restrictive topic filters, page settings, curriculum reference, and optional isolated delivery metadata. It intentionally does not store generated questions/answers or diagnostic item snapshots.

Export Blueprint / Import Blueprint

39. **Generate the assessment you want to keep.**
40. **Click Export Blueprint.** The app downloads a routine slim v1 blueprint JSON.
41. **Move or archive the JSON wherever you like.**
42. **Later, click Import Blueprint and choose the file.** EquationWright validates and applies the blueprint settings before rendering.

Saved blueprint library

Save current stores the current slim blueprint in browser local storage under a name you choose. Load reapplies it; Delete removes it.

Portable vs. browser-local

Saved blueprints are convenient but tied to that browser's local storage. Export Blueprint is the safer choice when you want a file you can move, back up, or keep long-term.

Copy share link

Copy share link puts a routine blueprint into the URL so opening the link recreates the same worksheet/quiz settings. If the blueprint carries a results submit URL, EquationWright shows the destination and asks for confirmation before the first post from that page load.

Legacy imports

v7.3 still reads the exact legacy UMWB_BLUEPRINT format for migration. New v7.3 routine writers use equationwright-blueprint-slim-v1. Unknown major formats are rejected instead of being silently treated as legacy data.

12. Smart Review and personal practice history

Start Smart Review (local)

Smart Review does not require an AI provider. It reads the structured quiz result and creates a fresh, bounded quiz aimed at the strongest evidence of what the learner needs next.

- **Perfect completed quiz:** creates a short challenge review with high variety.
- **Some unanswered but no incorrect answers:** creates a pacing/retry review.
- **No answered questions:** creates a short starter diagnostic.
- **Incorrect answers present:** creates remediation, prioritizing repeated mistake, topic, or unit signals when enough evidence exists.

The local Smart Review uses a fresh deterministic seed and removes class/submission delivery metadata from the generated review.

Personal practice history

43. **Export result JSON files from quizzes.**
44. **Open Personal practice history (local files).**
45. **Click Import practice history and choose one or more result JSON files.**
46. **Review learner groups, attempts, latest accuracy/completion, trends, topic trends, and recurring/resolved mistake patterns.**
47. **Use Start next Smart Review on a learner group.** EquationWright uses the most recent eligible current-version attempts, with up to a three-attempt history window for targeting.
48. **Click Clear loaded history when finished.**

Privacy behavior

Personal practice history stays in the current tab. It is not uploaded and is not automatically saved as a history database.

13. Class tools and result collection

Local file workflow — simplest and safest

49. Have students Export results (JSON).
50. Collect those files through your normal class workflow.
51. Open Class tools → Import class results (JSON files).
52. Choose Attempts or Unique aliases.
53. Review missed units, missed topics, and mistake patterns.
54. Export class summary, copy it for an AI tool, or use Analyze class with AI.

Attempts vs. Unique aliases

View	Meaning
Attempts	Counts every accepted nonduplicate result attempt. Useful for workload/activity.
Unique aliases	Uses the latest attempt for each aliased learner. Useful when you want one current contribution per learner.

Collector workflow

Class tools can attach a Class ID, Assignment ID, and teacher-owned Results submit URL to delivery. Students choose an alias when submitting. EquationWright does not provide the collector service; the collector belongs to the teacher/study group.

- **Submit URL:** a POST destination such as a teacher-owned web app/collector.
- **Load results from URL:** asks for a read URL, performs GET, and accepts a result object, an array, or a {results:[...]} wrapper containing mw-quiz-results-v1 records.
- **Class ID / Assignment ID:** optional labels that travel with delivery/results.

Use aliases, not real names

The interface explicitly instructs students to submit under an alias such as Student-17. The built-in class collector workflow is homework/practice collection, not secure testing.

If a collector fails

Students can always Export results (JSON) and send the file manually. The submission code also has a no-CORS retry path for collectors such as Google Apps Script where the browser may be unable to read the response.

14. AI tutor and local AI

Supported provider choices in the UI

- Anthropic (Claude)
- Google (Gemini)
- OpenRouter
- OpenAI-compatible custom endpoint — including local servers such as Ollama or LM Studio

Set up AI

55. **Open AI tutor (bring your own key).**
56. **Choose the Provider.**
57. **Enter a Model if you want to override the default.**
58. **For OpenAI-compatible, enter the Endpoint URL.** A local endpoint can be used without an API key when the server does not require one.
59. **Enter an API key for hosted providers.** The interface recommends a spend-capped throwaway key rather than a main account key.
60. **Choose Remember on this device only if you want browser storage to retain the key.** Unchecked means the key is forgotten when the tab closes.
61. **Take a quiz and choose Ask AI tutor, or use Analyze class with AI.**

What the AI receives and what it does

The browser calls the provider directly. The AI analyzes structured/labeled mistakes and proposes a bounded drill. EquationWright then validates/sanitizes that drill and generates/grades the mathematics locally.

Diagnostics and privacy

Diagnostic provenance is stripped from AI/provider transports, class aggregation, and result-submission transports. Deliberate local diagnostic and archival exports can retain it.

When AI is unavailable

EquationWright has a deterministic local Smart Review path. If an AI reply is unavailable, malformed, or rejected, the learner can still start local Smart Review rather than losing the next-practice workflow.

15. Advanced protocol, replay, diagnostics, and profiles

Advanced section

Most teachers and students do not need these controls. They are for integrations, bug reports, long-term evidence, curriculum mapping, and maintainers.

Export capabilities

Export capabilities downloads the machine-readable equationwright-capabilities-v1 description of the live engine, determinism protocol, catalog, units, generator IDs, topics, and supported public formats/features. Integrations should use this instead of scraping the HTML.

Result diagnostics

Level	Use
Off (normal/lean)	Normal student results. Recommended default.
Replay provenance	Adds the provenance needed for detailed diagnostic reproduction.
Full provenance + semantics	Adds fuller diagnostic provenance and semantic snapshots for advanced debugging.

Archival replay

Archival replay is intentionally different from a normal blueprint. An equationwright-blueprint-archive-v1 contains the authoritative slim blueprint plus ordered item snapshots and stable generator/distractor provenance.

Action	Meaning
Export archival replay	Create an explicit replay/debug/evidence package for the currently generated assessment.
Import archival replay	Runs strict replay. The archive must match the running assessment compatibility identity and catalog digest, then every archived item is reconstructed and compared.
Verify archive with current engine	Runs an explicit cross-fingerprint item-by-item comparison. This can tell you whether that particular archive still reconstructs, but it does not declare broad engine compatibility.

How to read replay results

- **exact / guaranteed:** strict replay succeeded under matching compatibility conditions.
- **incompatible:** the strict compatibility/catalog checks refused replay before claiming exactness.
- **mismatch / invalid archive:** the object or reconstructed item fields disagree.
- **presentation differences:** curriculum labels/course presentation are reported separately from assessed-math exactness.

Curriculum profiles

v7.3 ships a Generic profile and a VCCS/NVCC Mathematics profile. Profiles can change presentation—course groupings/order, labels/defaults, and descriptive crosswalks—without defining new mathematics.

- **Import curriculum profile:** loads one or more local JSON profile files after full validation.
- **Profiles may:** group/order known units, provide labels/defaults, and define descriptive crosswalks.
- **Profiles may not:** define generators, RNG behavior, answer validators, multiple-choice behavior, AI behavior, or mathematical content.

Stable generator identity

Published generator families have stable IDs in the form ewg.<unit-id>.<published-ordinal>. Runtime array positions and problem keys are diagnostic details rather than long-lived external identity.

16. Self-test and troubleshooting

Built-in Self-test

Self-test is mainly a teacher/developer/maintainer verification tool. It exercises live generators and reports structural problems rather than changing student work.

Control	Purpose
Scope	Current mode units or all units.
Samples per generator	1–200 generated samples per family.
Max error examples	1–50 detailed examples retained in the report.
Answer-capacity scan	Off, or 50/100/200 samples per generator to identify low choice capacity or generator execution errors.
Run self-test	Runs generator checks plus Statistics, Logic, and Differential Equations specialized suites.
Replay first failure	Recreates the stored unit/generator/run seed for the first captured failure.
Copy failure replay	Copies the exact replay payload for investigation.

Troubleshooting checklist

Problem	What to check
Formulas show raw LaTeX	Online edition: reconnect or configure Offline math

Problem	What to check
	engine with a MathJax tex-svg.js path/URL. Offline edition: use the approved offline HTML and reload.
Wrong number of questions	Check per-unit counts and Total Qs. Topic Clear does not disable a unit; set its count to 0.
A saved setup disappeared	The Saved blueprints library and working state use browser local storage. Use Export Blueprint for a portable backup.
Student cannot switch to Teacher	A locked Student link may have set a session PIN lock. Enter the teacher PIN or reopen outside that locked session.
Share link does not behave as expected	Recreate it from a current generated assessment. Avoid hand-editing the URL payload.
Results will not submit	Export JSON and send the file manually. Verify the collector URL and its permissions.
AI does not respond	Check provider/model/key/endpoint. For local servers, confirm the endpoint is reachable and allows browser requests. Local Smart Review remains available.
Archive strict replay says incompatible	That is expected when the assessment compatibility identity or catalog digest differs. Use Verify archive with current engine only as a diagnostic comparison.

17. Ways to use EquationWright

Use case	Recommended workflow
Daily warm-up	Use Quick Quiz (5) with a date/class seed. Students answer on their own device, then use Smart Review if needed.
Printable homework	Use Homework Practice (15), then Student Copy for delivery and Teacher Copy for your key/solutions.
Topic-specific review	Set all unwanted units to 0; within the target unit use Topics to narrow the concepts.
Common assessment	Create a fixed seed/settings package and distribute the share link or exported blueprint.
Parallel/make-up version	Keep counts/topics identical and use New quiz like this or a different seed.

Use case	Recommended workflow
Mixed-course diagnostic	Use Mixed + Quick Quiz (5) or a larger preset; seeded course round-robin samples multiple course families reproducibly.
Remediation loop	Take quiz → Review missed → Start Smart Review (local) → repeat.
Longer learner history	Export JSON after attempts → import into Personal practice history → Start next Smart Review.
Class review planning	Import class result JSON files → inspect missed units/topics/patterns → export/copy summary.
AI-assisted plan	Ask AI tutor or Analyze class with AI; let EquationWright validate the drill and continue grading locally.
Offline classroom/lab	Distribute the offline HTML. Formulas render without CDN access; use local file export/import for workflows that should stay disconnected.
Bug report / reproducibility	Use diagnostic results and an archival replay so a maintainer can reconstruct exact item and distractor provenance.
Curriculum remapping	Load a validated custom profile to present the same generator catalog under a different approved course grouping/labeling scheme.

18. Rebuilding from source

The final approved release package contains a canonical modular source ZIP, not just compiled HTML. It includes the source modules, profiles, schemas, registry/bindings, tests, pinned offline MathJax, and the deterministic Python build pipeline.

Where the builder is

- **source/EquationWright_v7_3_MODULAR_SOURCE.zip** — canonical development source snapshot.
- **build_release.py** — builds the online and offline distribution files.
- **release_gates.py** — runs the full release-gate workflow.
- **build/module-order.json** — defines the approved classic-script module order.
- **build/generator-bindings-v1.json** — binds stable generator IDs to source/runtime positions.
- **src/** — editable modular application source.
- **profiles/, schemas/, registry/, examples/** — public data/contracts and release inputs.

Basic commands

After extracting the modular source into a working folder:

```
python build_release.py
```

Build the generated dist/ artifacts.

```
python build_release.py --verify
```

Build and run the builder's verification path.

```
python build_release.py --release-gates
```

Run the complete release build/gate workflow used for release validation.

Do not hand-edit dist/

The release notes explicitly describe dist/ as generated output. Make changes in the modular source and rebuild so compatibility, registry/binding checks, manifests, online/offline packaging, and reproducibility checks remain meaningful.

Appendix A. Course and unit catalog

The built-in VCCS / NVCC Mathematics profile exposes 18 named mathematics courses plus Mixed. Mixed contains all 81 live units.

Course	Title	Units	Mode alias
MTH 133	Math for Health Professions	3	HP
MTH 154	Quantitative Reasoning	4	QR
MTH 161	Precalculus I	7	CA
MTH 162	Precalculus II: Trigonometry	7	TRIG
MTH 167	Precalculus with Trigonometry	16	PC
MTH 263	Calculus I	5	C1
MTH 264	Calculus II	5	C2
MTH 265	Calculus III	4	C3
MTH 266	Linear Algebra	5	LA
MTH 261	Applied Calculus I	3	AC1
MTH 262	Applied Calculus II	2	AC2
MTH 155	Statistical Reasoning	4	S155
MTH 245	Statistics I	4	S245
MTH 283	Probability and Statistics	4	P283
MTH 246	Statistics II	4	ST2
MTH 267	Differential Equations	5	DE
MTH 288	Discrete Mathematics	6	DM
MTH 289	Differential Equations of Mathematical Physics	7	D289

Mixed mode

Mixed: All Available Courses contains 81 units. The built-in Mixed preset allocator uses a seeded course round-robin so short presets sample multiple course families when available.

MTH 133 — Math for Health Professions

Unit ID	Unit label	Default count
HP_Conversion	Dimensional Analysis & Conversions	2
HP_Dosage	Dosage Calculations	2
HP_IVRates	IV Rates & Solutions	2

MTH 154 — Quantitative Reasoning

Unit ID	Unit label	Default count
QR_Proportion	Proportional Reasoning	2
QR_Finance	Financial Literacy	2
QR_Logic	Logic & Sets (Validity)	2
QR_Modeling	Linear Modeling	2

MTH 161 — Precalculus I

Unit ID	Unit label	Default count
CA_Linear	Linear Equations & Inequalities	2
CA_Quadratics	Quadratic Functions	3
CA_Polynomials	Polynomial Functions	2
CA_Rational	Rational Functions	2
CA_ExpLog	Exponential & Logarithmic	3
CA_Systems	Systems of Equations	2
CA_Functions	Function Operations	2

MTH 162 — Precalculus II: Trigonometry

Unit ID	Unit label	Default count
TRIG_Angles	Angle Conversions	2

Unit ID	Unit label	Default count
TRIG_UnitCircle	Unit Circle & Trig Values	3
TRIG_Identities	Trig Identities	3
TRIG_Equations	Solving Trig Equations	2
TRIG_Triangle	Law of Sines & Cosines	2
TRIG_Polar	Polar Coordinates	2
PC_Conics	Conic Sections	2

MTH 167 — Precalculus with Trigonometry

Unit ID	Unit label	Default count
CA_Linear	Linear Equations & Inequalities	2
CA_Quadratics	Quadratic Functions	3
CA_Polynomials	Polynomial Functions	2
CA_Rational	Rational Functions	2
CA_ExpLog	Exponential & Logarithmic	3
CA_Systems	Systems of Equations	2
CA_Functions	Function Operations	2
TRIG_Angles	Angle Conversions	2
TRIG_UnitCircle	Unit Circle & Trig Values	3
TRIG_Identities	Trig Identities	3
TRIG_Equations	Solving Trig Equations	2
TRIG_Triangle	Law of Sines & Cosines	2
TRIG_Polar	Polar Coordinates	2
PC_Conics	Conic Sections	2
PC_Sequences	Sequences & Series	2
PC_Vectors	Vectors (Intro)	2

MTH 263 — Calculus I

Unit ID	Unit label	Default count
C1_Limits	Limits	2
C1_Continuity	Continuity	1
C1_Derivatives	Derivatives	3
C1_Apps	Derivative Applications	2
C1_Integrals	Integrals (Intro)	3

MTH 264 — Calculus II

Unit ID	Unit label	Default count
C2_Tech	Integration Techniques	3
C2_Improper	Improper Integrals	2
C2_Series	Sequences & Series	3
C2_ParamPolar	Parametric / Polar	2
C2_AppsInt	Applications of Integration	2

MTH 265 — Calculus III

Unit ID	Unit label	Default count
C3_Vectors	Vectors & Geometry	3
C3_Partial	Partial Derivatives	3
C3_MultInt	Multiple Integrals	2
C3_VecCalc	Vector Calculus	2

MTH 266 — Linear Algebra

Unit ID	Unit label	Default count
LA_Systems	Solving Linear Systems	2

Unit ID	Unit label	Default count
LA_RowRed	Row Operations & RREF	2
LA_SolutionSets	Solution Sets & Rank	2
LA_MatrixAlg	Matrix Algebra & Determinants	2
LA_Eigen	Eigenvalues & Vector Spaces	2

MTH 261 — Applied Calculus I

Unit ID	Unit label	Default count
AC1_Marginal	Marginal Analysis	2
AC1_Elasticity	Elasticity of Demand	2
AC1_Finance	Continuous Growth & Finance	2

MTH 262 — Applied Calculus II

Unit ID	Unit label	Default count
AC2_IntApps	Integration Applications	2
AC2_Multivar	Multivariable Basics	2

MTH 155 — Statistical Reasoning

Unit ID	Unit label	Default count
S155_Descriptive	Descriptive Statistics	2
S155_Normal	Normal Distribution & z-Scores	3
S155_Correlation	Correlation & Causation	2
S155_Inference	Intervals & p-Values	2

MTH 245 — Statistics I

Unit ID	Unit label	Default count
S245_Probability	Probability Rules	2
S245_Binomial	Binomial Distribution	2
S245_Sampling	Sampling Distributions & CLT	2
S245_Inference	Confidence Intervals & Tests	2

MTH 283 — Probability and Statistics

Unit ID	Unit label	Default count
P283_Continuous	Continuous Random Variables	2
P283_ExpNormal	Exponential & Normal Models	2
P283_CLT_Est	CLT & Estimation	2
P283_LSQ	Least Squares & Correlation	2

MTH 246 — Statistics II

Unit ID	Unit label	Default count
ST2_ChiSquare	Chi-Square Tests	2
ST2_ANOVA	Analysis of Variance	2
ST2_Regression	Regression & r^2	2
ST2_Design	Experimental Design & Non-Parametrics	2

MTH 267 — Differential Equations

Unit ID	Unit label	Default count
DE_Basics	Classification & Qualitative Behavior	2
DE_Methods	Separable & Linear Methods	3
DE_AppsNum	Applications & Euler's Method	2

Unit ID	Unit label	Default count
DE_SecondOrder	Second-Order Linear Equations	2
DE_Laplace	Laplace Transforms	2

MTH 288 — Discrete Mathematics

Unit ID	Unit label	Default count
DM_Logic	Logic, Truth Tables & Quantifiers	2
DM_Sets	Sets & Set Operations	2
DM_Counting	Counting & Combinatorics	2
DM_RelFunc	Relations & Functions	2
DM_Graphs	Graphs & Trees	2
DM_ProofRec	Proofs, Growth & Recurrences	2

MTH 289 — Differential Equations of Mathematical Physics

Unit ID	Unit label	Default count
D289_FirstOrder	Typed First-Order IVP Review	2
D289_Systems	Systems of Differential Equations	2
D289_Series	Power-Series Solutions	2
D289_Fourier	Fourier Series	2
D289_Transforms	Laplace Transforms II	2
D289_PDE	PDE Classification & Separation	2
D289_BVP	Boundary-Value Eigenproblems	2

Appendix B. Quick reference

Choose this when you want...

Goal	Use
A 5-question interactive practice	Student → Quick Quiz (5)
A printable homework with solutions	Homework Practice (15)
A 20-question worksheet with answer key only	Unit Review (20)
The exact same quiz again	Retake this quiz
A fresh version with the same structure/settings	New quiz like this
A targeted local remediation quiz	Start Smart Review (local)
A portable setup file	Export Blueprint
A portable exact-debug/evidence package	Export archival replay
A shareable setup in a URL	Copy share link
Student-focused link with casual teacher-mode deterrence	Copy locked student link
A personal trend view	Import practice history
A class summary	Import class results → Attempts/Unique aliases
An AI recommendation	Ask AI tutor / Analyze class with AI
A full catalog/integration description	Export capabilities
A generator sanity check	Self-test
A fully bundled math renderer	EquationWright_v7_3_OFFLINE.html

Privacy and safety reminders

- Use learner aliases rather than real names in collector workflows.
- The Student-lock PIN is not exam security.
- Hosted AI calls go directly from the browser to the selected provider; use an appropriate spend-capped key.
- Result diagnostic provenance is off by default and is stripped from normal transports.
- Back up important assessments with exported blueprint/archive files rather than relying only on browser local storage.

Final release identity

Item	Approved v7.3 value
appVersion	v7.3_full_catalog_protocols
assessmentCompatibilityId	ewac:sha256: 2606c2ac96fc8ac65e9705e46f68a1ae 5d757d6a5454ca2ace74aee7b8f9dbe7
catalogDigest	sha256: 4cb338d5f4c2acb69fc9ec12cc6404e8 45180cc21264c12d8f1453ceccd3c1c2
Catalog	81 units / 424 generator families / 317 unit-scoped topics

Appendix C. Developer notes: build, register, and add new content

This appendix is for people maintaining or extending EquationWright. The safest rule is simple: edit the modular source, not the generated HTML. The source package declares `src/`, `profiles/`, `schemas/`, `registry/`, and `build/module-order.json` as build inputs; `dist/` is generated output.

Where to make changes

Goal	Main file(s)	What you change
Change an existing generator	<code>src/generators/<course>.js</code>	Edit the generator registration block, then update its binding digest intentionally.
Add a generator to an existing unit	<code>src/generators/<course>.js</code> + <code>registry/</code> + <code>build/generator-bindings-v1.json</code>	Append a <code>registerGen(...)</code> block and assign a new stable generator ID.
Add a unit	<code>src/catalog/units.js</code> + generator source + <code>profiles/</code>	Add the unit definition, generators, and profile course membership/default count.
Add a course/profile mapping	<code>profiles/*.json</code>	Add or revise the course object, <code>unitOrder</code> , <code>defaultCounts</code> , <code>aliases</code> , and <code>crosswalks</code> .
Add a new source module	<code>build/module-order.json</code> + <code>build_release.py</code>	Add it in deterministic order and deliberately update the builder's exact module-count expectation.
Create release files	<code>build_release.py</code>	Generate <code>dist/</code> online/offline HTML, capability manifests, <code>schemas/examples</code> , and build manifest.

1. Rebuild the application

1. Extract `source/EquationWright_v7_3_MODULAR_SOURCE.zip` into its own working folder.
2. Open a terminal in that folder. Do not edit the generated files in `dist/` as your source of truth.
3. Run the normal build. It creates the online and offline single-file applications plus the public manifests and release contracts.

```
python build_release.py
```

For a reproducibility check, ask the builder to make a second independent build and compare the generated artifacts byte-for-byte.

```
python build_release.py --verify
```

Before a release, run the complete release gate. This rebuilds, checks reproducibility, and then runs the Stage 3/Stage 4, Pass 4C, math, offline, replay, and UI test scripts.

```
python build_release.py --release-gates
```

Build-environment note: the v7.3 builder generates the live capability manifest through Playwright and launches Chromium from /usr/bin/chromium. The release gate also imports jsonschema. A clean development machine therefore needs Python, Playwright, Chromium, and jsonschema available for the full build/release path.

2. What "registering a generator" means

EquationWright has two related registration layers. Both matter.

- Runtime registration: a source generator is added with `registerGen('UNIT_ID', (rng)=>{ ... })`. The order of those calls becomes the runtime index inside that unit.
- Stable public registration: `registry/EquationWright_Generator_Registry_v1.json` assigns the published `generatorId`, `unit`, `ordinal`, `runtime slot`, `topic information`, and `observed capability data`.
- Source binding: `build/generator-bindings-v1.json` binds that stable generator ID to its `unit/runtime slot`, `source module`, and `SHA-256 of the exact registration block`.

Why this is strict: v7.3 intentionally freezes the 424 published generator identities. Reordering two existing `registerGen` calls, moving a family to another slot, or editing a bound registration block without deliberately updating its binding is rejected by the builder. This protects saved archives, replay provenance, and stable generator IDs.

3. Add a new generator to an existing unit

1. Pick the existing unit ID in `src/catalog/units.js`, then open that course's file in `src/generators/`.
2. Append the new `registerGen(...)` block after the existing registrations for that unit. Do not insert it in the middle of the published v7.3 sequence.
3. Keep the generator deterministic. Use the provided `rng/helpers`; do not use `Math.random()`, current time, network responses, or other uncontrolled state.
4. Return the normal problem fields used by the catalog: `q`, `a`, `steps`, `traps`, `vec`, and a stable key. Add returned meta `{ topicId, topicLabel }` when you want an explicit topic instead of relying on derivation.
5. Give the new family a new stable ID in `registry/EquationWright_Generator_Registry_v1.json`. Use the next published ordinal for that unit and the new runtime index; never recycle an older stable ID for different math.
6. Add the matching row to `build/generator-bindings-v1.json` with the same `generatorId/unit/runtime index`, the `source module`, and the `SHA-256 of the exact registerGen block`.
7. Because the v7.3 builder deliberately asserts 424 registrations/bindings, a future catalog release that adds families must intentionally update those expected-count checks as part of the versioned change.

8. Build and run the release gates. Treat failures as useful guardrails rather than bypassing them.

Minimal generator shape

```
registerGen('MY_Unit', (rng)=>{
  const n = pick(rng, [2,3,4]);
  return {
    q: `Example question using ${n}`,
    a: `${n*n}`,
    steps: ['Work the example deterministically.'],
    traps: [{ans: `${n+n}`, why: 'Example misconception.'}],
    vec: [unitIndex('MY_Unit'), 1, n],
    key: 'my_unique_problem_key',
    meta: { topicId: 'example', topicLabel: 'Example' }
  };
});
```

The snippet shows the shape, not a complete new feature. A production generator still needs mathematically correct answer logic, useful worked steps, sensible distractors, topic/capacity behavior, and release-gate coverage.

4. Add a new unit or course

- New unit: add { course, id, label, defaultCount } to src/catalog/units.js, then add its generator registrations.
- Profiles: add the unit to unitOrder/defaultCounts. A new course needs courseId, title/code, unique aliases, unitOrder, and defaults.
- Crosswalks: update only when the mapping really changes. Existing course membership/order/default or crosswalk-meaning changes require a major profile-version change.
- New source module: add it to build/module-order.json in deterministic order. v7.3 expects exactly 48 modules, so adding one also requires an intentional build-tool/version update.
- Update the stable generator registry and source-binding manifest for every new family, then rebuild and run the full release gates.

5. Compatibility rule to remember

Assessment behavior is fingerprinted. Determinism-critical source, registry/order, algorithms, and assessment-relevant profile mappings can change assessmentCompatibilityId. That is expected for real assessed-content changes; never preserve the old ID by hard-coding it.

Presentation-only labels/text are different, but course/unit/default mappings participate in the assessment identity.

Safe extension checklist

- Work from modular source; never treat generated dist/ HTML as the authoritative source.
- Append new families; keep published order/IDs stable and add matching registry/binding records.
- Update units, profiles, module order, and expected counts only when the new content requires them.

End of manual